

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I JAGADISH.V(1925213520(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Handling Seasonal Traffic Spikes with Auto-Scaling and Load Balancing

During seasonal sales, website traffic can increase suddenly. Cloud platforms use **auto-scaling** and **load balancing** to manage these traffic spikes efficiently.

Auto-Scaling

- Automatically increases the number of servers or virtual machines when demand rises.
- Reduces resources when traffic decreases.
- Ensures applications have sufficient computing power during peak periods.

Load Balancing

- Distributes incoming user requests across multiple servers.
- Prevents any single server from becoming overloaded.
- Maintains application responsiveness and availability.

Reducing Downtime and Improving User Experience

- Additional instances are launched automatically during heavy traffic.
- Traffic is evenly distributed among healthy servers.
- If one server fails, requests are redirected to other available servers.
- Users experience faster page loads and uninterrupted service.

2. Secure Cloud Storage for Large-Scale Backups

Durability and Data Redundancy

Cloud storage providers ensure data durability through:

- Multiple copies of data stored across different disks.
- Replication across multiple availability zones or regions.
- Automatic recovery from hardware failures.

This ensures backup data remains available even if individual storage devices fail.

Encryption for Data Protection

Cloud providers use:

- **Encryption at Rest:** Protects stored backup files.
- **Encryption in Transit:** Protects data while being transferred over networks.

Even if data is intercepted, it remains unreadable without the encryption key.

Access Control Mechanisms

Security is enhanced using:

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Identity and Access Management (IAM) policies

These controls ensure only authorized users can access backup data.

3. Global Deployment Using Content Delivery Networks (CDNs)

What is a CDN?

A **Content Delivery Network (CDN)** is a distributed network of servers that caches content closer to end users.

How CDNs Improve Performance

- Frequently accessed content is stored at edge locations.
- Users retrieve data from nearby servers rather than distant origin servers.
- Reduces latency and improves loading speed.

Role of Edge Servers

Edge servers:

- Store cached copies of web pages, images, videos, and application content.
- Reduce the distance data must travel.
- Decrease network congestion.
- Improve response times for users worldwide.

Supporting Global Availability

Cloud providers maintain:

- Multiple geographic regions
- Availability zones
- Global load balancers
- Automated failover mechanisms

If one region experiences issues, traffic can be redirected to another region.

Real-Time Streaming Application Example

Consider a video streaming platform such as Netflix.

Without a CDN:

- Users far from the origin server experience buffering and delays.

With a CDN:

- Videos are cached at edge servers worldwide.
- Users receive content from the nearest location.
- Streaming quality improves and buffering decreases.

4. Cloud Compliance and Security for a Financial Institution

Regulatory Compliance and Auditing

Financial institutions handle highly sensitive customer and transaction data. Cloud providers support compliance by offering certifications and security controls that help organizations meet regulatory requirements.

Examples include:

- PCI-DSS for payment card security
- ISO 27001 for information security management
- SOC audits for operational security
- Financial industry compliance frameworks

Cloud providers also provide audit reports and compliance tools that help organizations demonstrate adherence to regulations.

Role of Logging

Logging records all user and system activities, including:

- Login attempts
- Data access events
- Configuration changes
- Financial transactions

Logs create an audit trail that helps detect suspicious activities and supports compliance audits.

Role of Monitoring

Monitoring continuously tracks:

- System performance
- Unauthorized access attempts
- Network activity
- Security incidents

Real-time monitoring allows rapid detection and response to threats.

5. Containerization and Kubernetes for Application Deployment

Performance and Resource Usage

Containers package only the application and its dependencies, making them lightweight and efficient.

Benefits:

- Faster startup times
- Lower memory consumption
- Better CPU utilization
- Higher application density on the same hardware

Virtual machines require separate operating systems, consuming more memory, storage, and processing resources.

Kubernetes and Orchestration

Kubernetes automates the deployment, management, and scaling of containers.

Key functions include:

- Automated deployment

- Load balancing
- Self-healing of failed containers
- Automatic scaling based on demand
- Rolling updates with minimal downtime

Simplified Scaling

When application traffic increases:

1. Kubernetes detects higher resource usage.
2. Additional container instances are created automatically.
3. Traffic is distributed among available containers.
4. Resources are reduced when demand decreases.

Portability Across Cloud Environments

Containers package applications consistently, allowing them to run on:

- Private clouds
- Public clouds
- Hybrid cloud environments
- On-premises infrastructure

This reduces vendor lock-in and simplifies migration between cloud providers.

Example: Microservices-Based Application

Online Food Delivery System

Microservices may include:

- User Service
- Restaurant Service
- Order Service
- Payment Service
- Notification Service

Each service runs in its own container.